

WESTPORT, New Zealand

WMO: 935150

Lat: 41.7406S

Lon: 171.5797E

Elev: 15

StdP: 14.69

Time Zone: 12.00 (NZL)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	35.4	37.3	25.9	19.9	43.9	28.9	22.9	43.1	33.4	52.9	28.6	53.4	7.6	160	0.547

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	12.5	72.1	64.7	70.4	63.9	69.0	63.1	66.9	69.8	65.6	68.5	64.5	67.5	11.1	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.7	95.0	68.3	64.5	90.8	67.2	63.2	86.8	66.1	31.5	69.8	30.5	68.5	29.6	67.7	71.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.6	24.1	21.0	33.0	75.5	1.5	2.5	31.9	77.3	31.0	78.8	30.2	80.2	29.1	82.0		
(5)			30.7	68.3	1.6	1.6	29.5	69.4	28.6	70.3	27.8	71.2	26.6	72.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	54.7	61.4	62.0	59.9	56.1	52.8	49.2	47.9	48.8	51.2	52.9	55.2	58.9	(6)
(7)		DBStd	5.90	3.63	3.39	3.47	3.34	3.92	3.71	3.43	3.27	3.06	3.07	3.37	3.67	(7)
(8)		HDD50	254	0	0	0	2	16	58	84	61	22	9	2	0	(8)
(9)		HDD65	3798	120	93	161	266	377	473	530	501	415	376	295	191	(9)
(10)		CDD50	1955	354	336	308	186	105	36	18	25	56	98	157	276	(10)
(11)		CDD65	24	9	9	4	0	0	0	0	0	0	0	0	2	(11)
(12)		CDH74	22	11	7	3	1	0	0	0	0	0	0	0	0	(12)
(13)		CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	8.7	9.0	8.2	7.9	7.9	8.1	8.4	8.3	8.3	9.4	10.1	10.0	9.4	(14)
(15)	Precipitation	PrecAvg	77.6	5.9	4.3	5.1	6.1	7.1	8.2	6.1	6.5	6.8	7.9	5.8	7.6	(15)
(16)		PrecMax	99.1	10.8	8.7	9.0	12.2	14.2	12.5	12.7	9.6	13.4	13.5	11.6	13.0	(16)
(17)		PrecMin	60.9	2.1	1.2	0.7	1.7	2.6	3.5	1.3	2.5	2.6	4.4	1.2	3.0	(17)
(18)		PrecStd	9.2	2.0	1.7	2.3	3.0	2.8	2.5	2.8	1.8	2.8	2.5	3.1	2.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	75.5	74.7	72.4	67.7	65.4	60.2	58.9	60.2	62.2	63.8	68.4	72.0	(19)
(20)			MCWB	67.1	65.8	62.9	60.8	57.7	54.9	52.0	52.2	54.6	57.0	60.8	63.7	(20)
(21)		2%	DB	72.2	72.1	70.0	65.3	62.5	58.4	57.1	57.7	59.5	61.8	65.3	69.4	(21)
(22)			MCWB	65.1	65.3	63.1	59.8	58.3	54.8	52.1	52.2	54.3	56.8	59.2	62.9	(22)
(23)		5%	DB	70.1	70.5	68.3	64.0	61.0	57.3	55.8	56.3	58.2	60.3	63.3	67.3	(23)
(24)			MCWB	63.9	64.4	62.6	59.3	57.3	54.0	51.5	51.9	53.8	55.8	58.1	61.6	(24)
(25)		10%	DB	68.4	68.9	66.9	62.7	59.7	56.1	54.6	55.1	57.0	59.0	61.7	65.5	(25)
(26)			MCWB	63.0	63.5	61.9	58.6	56.4	53.2	50.7	51.4	53.4	54.8	57.0	60.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	68.8	68.7	67.0	64.0	61.5	58.1	56.2	55.8	57.5	59.9	62.2	65.7	(27)
(28)			MCDWB	73.1	71.6	69.4	65.7	62.4	58.7	56.9	56.8	59.0	61.8	65.7	69.2	(28)
(29)		2%	WB	66.8	67.2	65.5	62.2	59.9	56.5	54.6	54.5	56.5	58.3	60.7	64.1	(29)
(30)			MCDWB	69.8	70.2	67.9	63.5	61.2	57.5	55.8	55.9	57.9	60.2	63.5	66.7	(30)
(31)		5%	WB	65.4	65.8	64.2	61.0	58.4	55.3	53.0	53.2	55.6	57.1	59.5	63.0	(31)
(32)			MCDWB	68.3	69.0	66.9	62.6	60.1	56.6	54.7	55.1	57.1	58.9	61.9	65.5	(32)
(33)		10%	WB	64.1	64.5	62.9	59.5	56.9	53.7	51.3	52.1	54.5	56.1	58.4	62.0	(33)
(34)			MCDWB	67.3	67.9	65.8	61.7	59.2	55.5	53.7	54.4	56.4	58.1	60.7	64.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	11.7	12.5	12.5	12.3	11.9	11.5	12.2	12.0	11.4	11.2	11.4	10.8	(35)
(36)			MCDBR	13.5	13.7	13.7	13.2	11.7	11.2	12.1	12.6	12.3	12.1	12.6	12.6	(36)
(37)			MCWBR	8.4	8.4	8.7	9.2	8.6	8.7	8.8	8.9	8.9	8.7	8.2	8.1	(37)
(38)		5% WB	MCDBR	10.8	11.2	10.9	9.6	9.1	8.7	9.8	10.1	9.5	9.4	9.7	9.6	(38)
(39)			MCWBR	7.6	8.2	7.8	8.2	7.9	8.2	8.7	8.2	8.0	8.0	7.5	7.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.353	0.343	0.326	0.314	0.302	0.299	0.291	0.306	0.334	0.343	0.348	0.345	(40)	
(41)		taud	2.534	2.582	2.635	2.650	2.654	2.628	2.650	2.624	2.522	2.515	2.510	2.567	(41)	
(42)		Ebn at Noon	309	304	296	278	260	249	262	277	287	300	308	313	(42)	
(43)		Edh at Noon	34	31	27	23	19	18	19	23	30	33	35	33	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1999	1781	1355	891	577	446	530	757	1088	1460	1863	1968	(44)	
(45)		RadStd	141	141	79	81	43	37	45	39	90	121	177	158	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+7
(47) Regional (4 neighbors)	+0.54	N/A	N/A	N/A	N/A	N/A	-65	-162	+131	N/A

Nomenclature: See separate page